Employee Workforce Analytics Dashboard

Power BI Project Report

Project Type	Business Intelligence / Data Analytics
Primary Tool	Microsoft Power BI
Dataset Size	3,000 synthetic employee records
Purpose	Portfolio documentation for GitHub

Prepared as a portfolio report. Update any descriptions if your final Power BI report differs from this documented version.

1. Project Overview

The Employee Workforce Analytics Dashboard is an interactive business intelligence project built in Microsoft Power BI. It explores workforce distribution, hiring patterns, attrition, employee satisfaction, income, experience, and demographics through KPI cards, charts, slicers, tables, and matrices.

2. Problem Statement

Organizations need a clear view of workforce composition and patterns in recruitment, attrition, satisfaction, and compensation. Reviewing raw employee records manually can make comparisons difficult. This dashboard presents workforce information interactively so users can compare departments, locations, job roles, and years.

3. Objectives

- Summarize the workforce using key performance indicators (KPIs).
- Compare employee counts across departments and locations.
- Explore hiring patterns by month and day.
- Analyze attrition by employment type and job role.
- Compare satisfaction and average monthly income across departments.
- Explore education and gender distribution.
- Enable interactive analysis using Year and Location slicers.

4. Dataset Description

The project uses a synthetic dataset containing 3,000 employee records. Fields include Employee ID, Hire Date, Year, Department, Job Role, Location, Gender, Education, Employment Type, Age, Total Experience, Years at Company, Monthly Income, Training Hours, Job Satisfaction, Work Life Balance, Performance Rating, Overtime, Attrition, and calendar fields. This dataset is for learning and portfolio demonstration; it is not real company data.

5. Tools and Technologies

- Microsoft Power BI Desktop — report development and visualization.
- Power Query — data preparation and transformation.
- DAX — measures and KPI calculations.
- Excel — source data format and initial inspection.
- Data visualization principles — layout, chart selection, filtering, and storytelling.

6. Data Preparation

The workflow includes importing the Excel dataset, checking column names and data types, converting Hire Date to the Date data type, validating numeric fields, and reviewing categorical values. Sort Month Name by Month Sort and Day by Day Sort so visuals follow calendar order rather than alphabetical order.

7. Data Model and DAX Measures

The employee dataset is the main analytical table. Example measures below assume the table is named 'Employee Data'; replace that name if your Power BI table differs.

```
Total Employees = DISTINCTCOUNT('Employee Data'[Employee ID])

Total Attrition = CALCULATE(DISTINCTCOUNT('Employee Data'[Employee ID]), 'Employee
Data'[Attrition] = "Yes")

Attrition Rate = DIVIDE([Total Attrition], [Total Employees], 0)

Average Monthly Income = AVERAGE('Employee Data'[Monthly Income])

Average Experience = AVERAGE('Employee Data'[Total Experience (Years)])
```

Format Attrition Rate as a percentage.

8. Dashboard Components

KPI cards show Total Employees, Average Monthly Income, Average Experience, and Total Attrition. Department and location charts show workforce distribution. A month-based chart explores hiring patterns. Attrition charts compare employment types and job roles. A satisfaction chart compares average job satisfaction by department. Tables and matrices show job-role analysis, education and gender distribution, and year-wise department counts. Year and Location slicers filter the report.

9. Key Findings

The dataset contains 3,000 employee records and 365 attrition cases, corresponding to an overall attrition rate of approximately 12.2% (365 ÷ 3,000). In the current report, IT has the highest employee count and HR has the lowest among the departments shown. These findings describe this synthetic dataset only. Validate the current Year and Location filters before interpreting chart results.

10. How to Use the Dashboard

1. Open the Power BI report.
2. Use the Year slicer to select one or more years.
3. Use the Location slicer to filter by location.
4. Review KPI cards in the selected filter context.
5. Select chart elements to cross-filter other visuals where enabled.
6. Use tables and matrices for detailed comparisons.
7. Clear slicers to return to the overall view.

11. Limitations

The dataset is synthetic and may not represent real workforce behavior. The dashboard is descriptive and does not establish why employees leave or predict future attrition. Results depend on data quality, measure definitions, and slicer selections. It should not be used to make employment decisions about real individuals.

12. Future Enhancements

- Add attrition trends by year and month.
- Add attrition-rate breakdowns by department and job role.
- Create drill-through pages for department-level details.
- Add KPI definitions and tooltips.
- Document data refresh and validation steps.

13. Conclusion

This project demonstrates a practical business intelligence workflow: importing data, preparing fields, creating DAX measures, and designing an interactive dashboard. It provides experience with Power BI, Power Query, data modeling concepts, and visual storytelling.

14. Suggested GitHub Repository Contents

• README.md — project summary, features, tools, and instructions.
• Employee_Workforce_Analytics_Dashboard.pbix — Power BI report, if you choose to upload it.
• Employee_Workforce_Analytics_Dataset.xlsx — dataset, if redistribution is permitted.
• screenshots/dashboard-overview.png — dashboard preview image.
• Employee_Workforce_Analytics_Project_Report.pdf — detailed report.